Firmware Upload Guide

Step 1. Install Arduino IDE

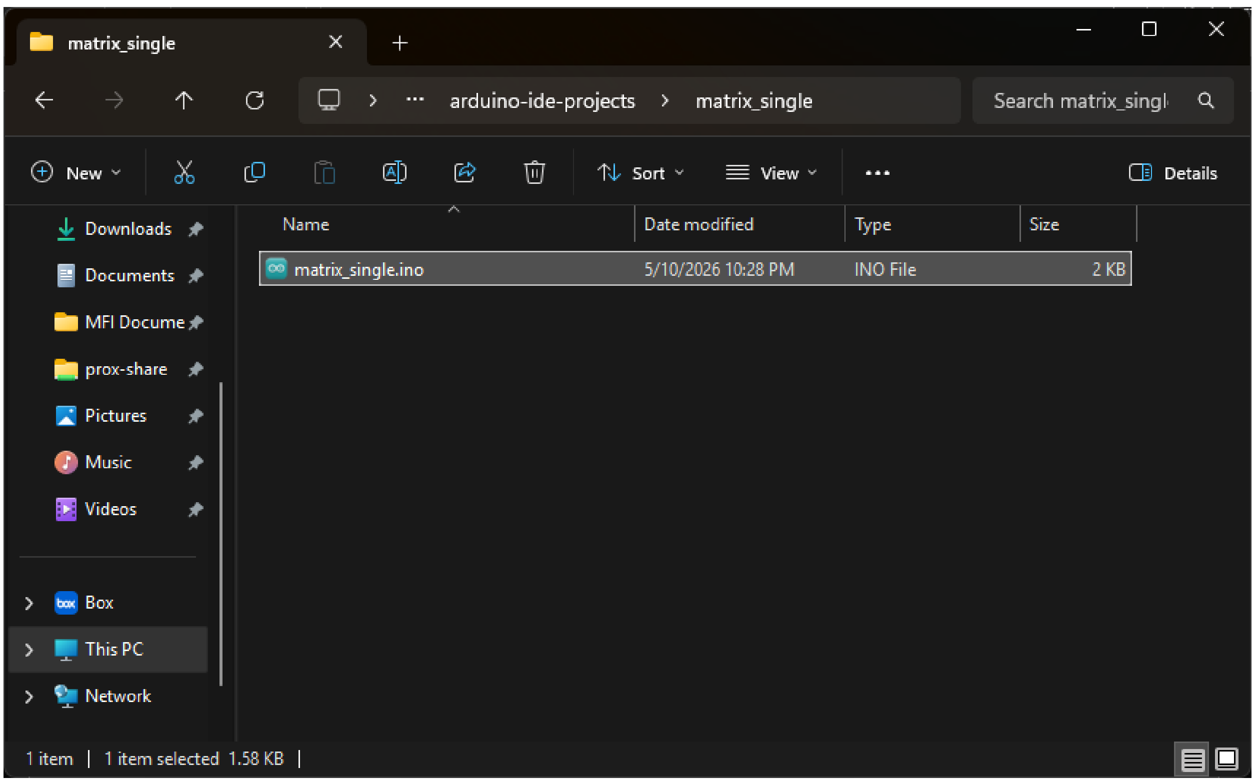
- Follow steps here: https://www.arduino.cc/en/software/

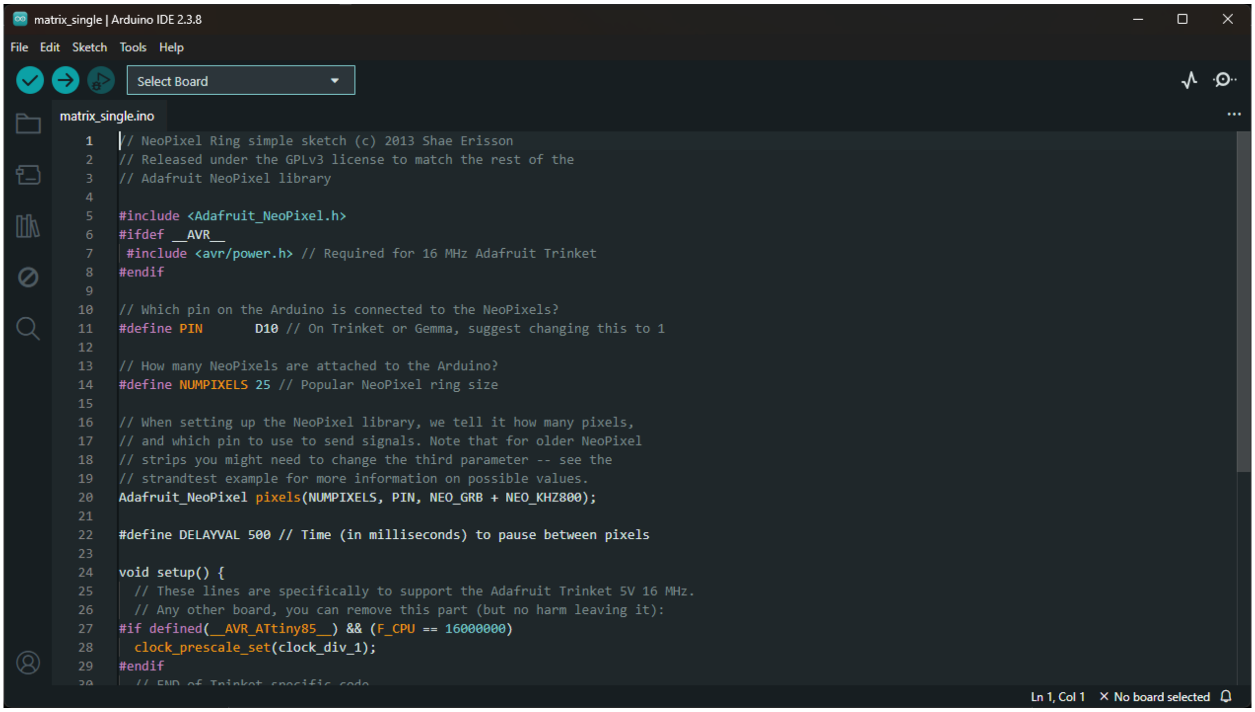
Step 2. Setup Arduino IDE for ESP 32

- Follow steps here: https://developer.espressif.com/blog/2025/10/arduino-get-started/

Step 3. Open a project

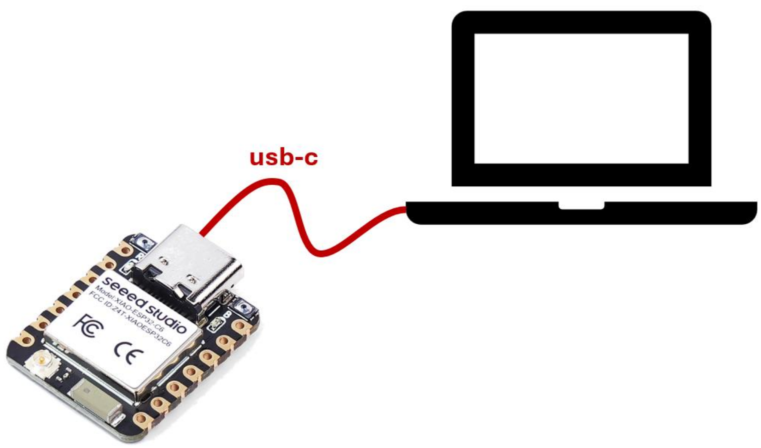
- Open .ino file in the projects



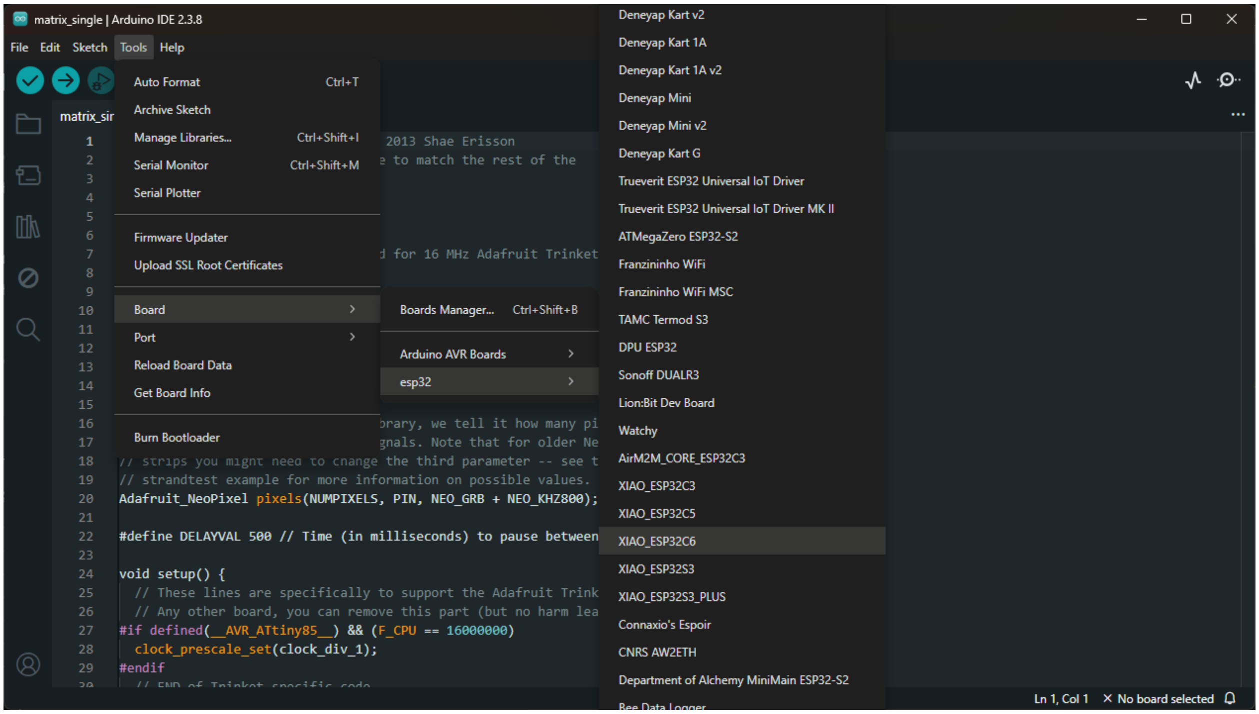


Step 4. Setup Board and the Port

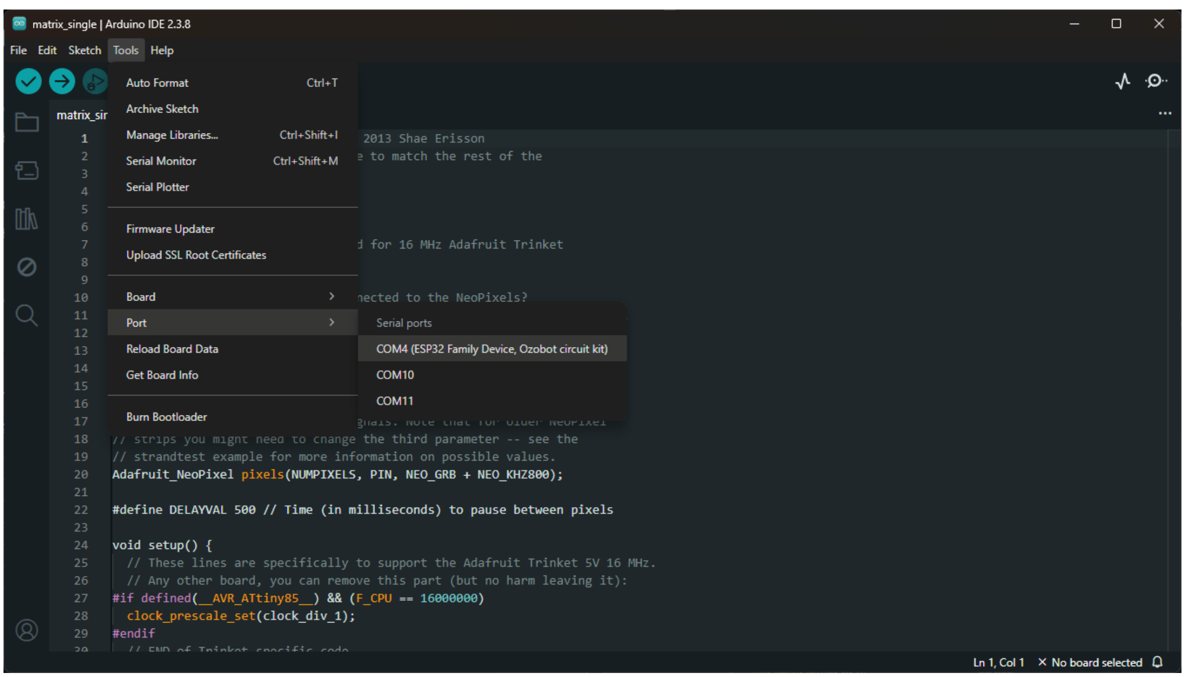
- Connect a USB-C cable from PC to the microcontroller board directly.



- Select the right board from **Tools -> Boards -> esp32 -> XIAO_ESP32C6**

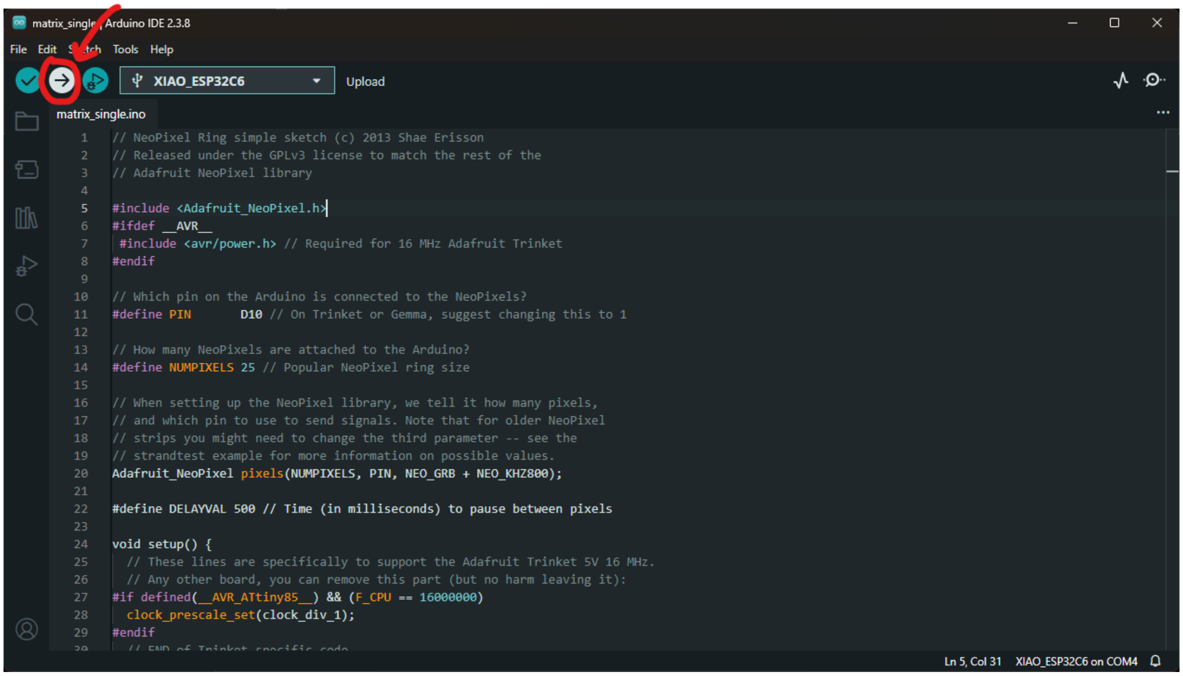


- Select the right port from **Tools -> Port -> COM3** (or higher). Choose the port that has the description “ESP32 Family Device....”

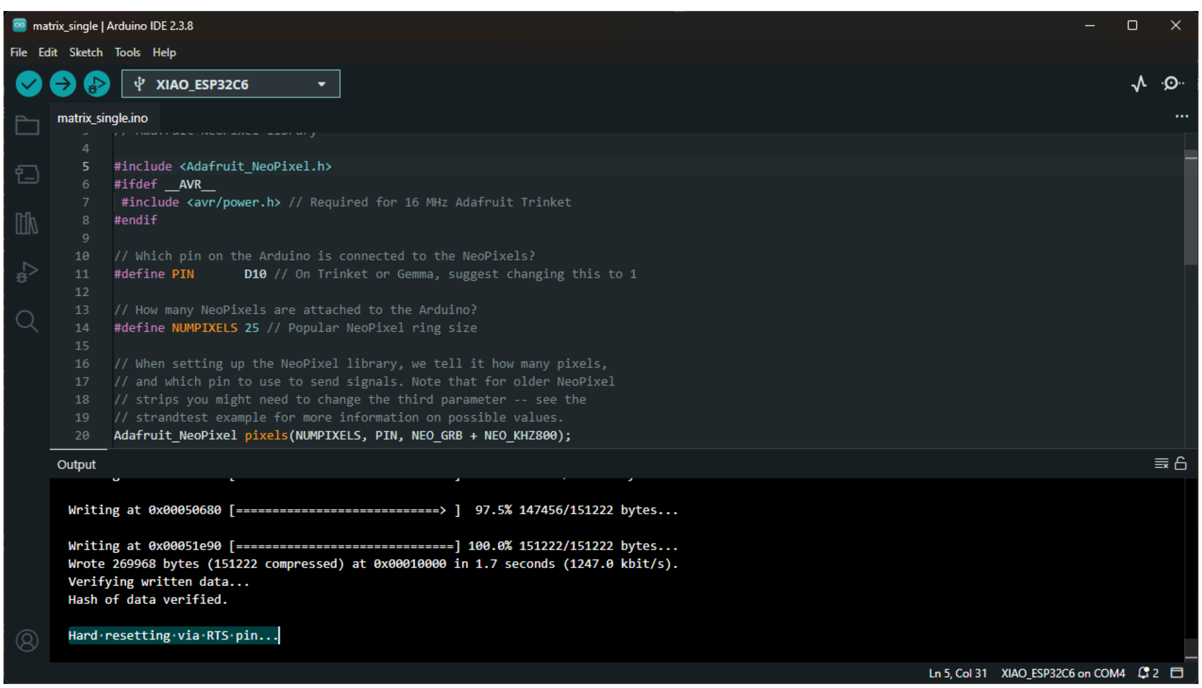


Step 5. Upload and Test

- Click Upload icon on top left.



- The program will compile and upload the compiled binary on the microcontroller board. It may take 5-10 min. Once uploaded the Output screen should print *"Hard resetting via RTS pin...."*



Step 6. Success

- Success! The lamp should light up!
- Unplug the USB